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PROPELLANT CHARGE, PROPELLANT CHARGE ASSEMBLY, AND CARTRIDGE AMMUNITION

Abstract

A propellant charge for driving a projectile, comprising a shell, which outwardly delimits a propellant chamber. At least two propellant charge molded bodies are arranged in the propellant chamber. Each of the propellant charge molded bodies has holes extending next to one another, and at least one of the propellant charge molded bodies has at least 70 respective holes extending next to one another. An ammunition comprising a projectile and such a propellant charge is also provided.

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Background/Summary

[0001] This nonprovisional application is a continuation of International Application No. PCT/EP2023/077781, which was filed on Oct. 6, 2023, and which claims priority to German Patent Application No. 10 2022 131 077.9, which was filed in Germany on November 24, 2022, and which are both herein incorporated by reference.

BACKGROUND OF THE INVENTION

Field of the Invention

[0002] The invention relates to a propellant charge for driving a projectile. The invention also relates to a propellant charge assembly and relates to an ammunition having the features of the other independent claim.

Description of the Background Art

[0003] In military technology, a propellant charge is used to drive a projectile through a gun barrel, and specifically by generating gas pressure. After ignition, the propellant charge burns, wherein this combustion burn-up (combustion gases) creates a high gas pressure “behind the projectile”, which drives the projectile toward the muzzle.

[0004] The performance of projectiles or ammunition is determined by the maximum permissible gas pressure in the gun barrel as well as by the charge mass, the chemistry and the grain geometry of the propellant powder. The current developments of projectiles, e.g., kinetic energy projectiles or KE projectiles (penetrators), are reaching their limits, because the propellant chamber volume and thus the possible available chemical energy of the propellant powder are limited.

[0005] Furthermore, modern high-performance weapons are limited in their maximum gas pressure by the material properties and the barrel geometry of the weapon. An enlargement of the propellant chamber, for example of its diameter, would be achieved at the cost of a considerable increase in the mass of the weapon. An extension of the propellant chamber would reduce the effectiveness of the drive.

[0006] Propellant powder geometries having 7, 19 or 37 holes are known from the state of the art. This makes it possible to create a surface area that increases with the degree of combustion of the propellant powder geometries. By increasing the surface area during combustion (progressive combustion behavior), the amount of gas produced is constantly increased, so that the projectile is accelerated over a longer period of time or a longer distance. However, larger propellant powder geometries with many holes reduce the fill level of propellant powder in the propellant charge or cartridge.

SUMMARY OF THE INVENTION

[0007] It is therefore an object of the invention to provide a propellant charge in which the performance of a projectile or ammunition can be increased and the maximum necessary gas pressure can be kept comparatively low.

[0008] The propellant charge is configured and/or intended to drive a projectile through a gun barrel. The propellant charge has a shell which outwardly delimits a propellant chamber and at least two, or more than two, propellant charge molded bodies arranged in the propellant chamber. The propellant charge molded bodies each have a plurality of holes extending next to one another. The holes may preferably extend parallel to one another. At least one of the propellant charge molded bodies, preferably two, or more than two, propellant charge molded bodies each have at least 60 or at least 70 holes extending next to one another.

[0009] The proposed design has the advantage that such propellant charge molded bodies (multi-hole molded bodies) burn with a multiple of the usual 19-hole surface increase. Due to the even more progressive combustion behavior (sharp increase in surface area during combustion), the gas mass that is produced per unit of time is constantly increased. This causes the projectile to accelerate over a longer period of time or over a longer distance, because the maximum gas pressure occurs at a later point in time or when the projectile has traveled a longer distance. The increased gas mass together with the gas release characteristics ensure a significantly increased output power of the projectile with comparatively moderate maximum gas pressure values.

[0010] Preferably, two, or more than two, propellant charge molded bodies can each have at least 60 or at least 70 holes extending next to one another. Due to the increased gas mass, the output power of the projectile can be increased even further. Further preferably, half of the propellant charge molded bodies, three quarters of the propellant charge molded bodies, or all of the propellant charge molded bodies can each be provided with at least 60 or at least 70 holes. This further improves the projectile's output power.

[0011] The holes or hole channels of a propellant charge molded body together form a perforation of the propellant charge molded body. The holes or hole channels can be designed as blind holes or as through holes.

[0012] The at least one propellant charge molded body can have at least 100, preferably at least 200, more preferably at least 300, even more preferably at least 400 holes extending next to one another. With such a multi-hole molded body, the gas mass and thus the output power of the projectile can be increased even further. This allows a particularly progressive combustion behavior to be achieved.

[0013] Preferably, two or more than two propellant charge molded bodies can each have a number of holes as stated above. Further preferably, half of the propellant charge molded bodies, three quarters of the propellant charge molded bodies, or all of the propellant charge molded bodies can each have a number of holes as stated above.

[0014] During tests, a propellant charge molded body with 470 holes extending next to one another, in particular holes extending parallel to one another, has proven to be particularly advantageous.

[0015] The holes in the propellant charge molded body can be made by drilling. However, this is not economically feasible, particularly for propellant charge molded bodies having a high number of holes, e.g., having 30 or more holes.

[0016] The propellant charge molded bodies proposed here can be extruded or produced by extrusion. The holes in the propellant charge molded body are formed directly during extrusion.

[0017] Specifically, the propellant charge molded bodies can be extruded in a quasi-continuous method. The still pasty ("dough-like") propellant charge mass is subjected to high pressure pressed through a die assembly in which the propellant charge molded body mass is separated into many individual strands and the holes are formed, wherein, after the die assembly, the individual strands are (again) pressed together into a strand of propellant charge molded body mass through a funnel-shaped channel. The strand can have any cross section, preferably a circular cross section.

[0018] Portions of the desired length can be cut from this strand (1st cut for trimming the strand to length), wherein the portions themselves directly form the propellant charge molded bodies (finished in the mold) or form the starting point for later propellant charge molded bodies. In the latter case (starting point for later propellant charge molded bodies), it is possible that the portion can be cut parallel or diagonally to the strand axis (2nd cut to "cut into shape"), so that a segment (e.g., a "pie slice" or a part thereof) is created as a propellant charge body (finished in the mold).

[0019] The die assembly can comprise a stable steel disk with a high number of holes (similar to a "meat grinder"; the number of holes can correspond to the desired number of holes in the propellant charge molded bodies, or can be lower or higher). Many very thin needles are attached to or in the steel disk (the number of needles corresponds to the number of holes in the later propellant charge molded body), wherein each needle forms a hole in the strand (later hole in the

propellant charge molded body). The needles are designed with such lengths that they extend from the steel disk or from within the steel disk to beyond the funnel-shaped channel in which the propellant charge molded body mass is pressed together to form a strand. The needles “float”, so to speak, in the compressed propellant charge molded body mass and leave behind the holes (holes in the propellant charge molded bodies).

[0020] Advantageously, the propellant charge molded bodies can be arranged or layered on top of one another in the chamber in one, two or more layers, wherein at least two propellant charge molded bodies each form a layer, wherein at least in one of the layers, preferably in two, a plurality of or all of the layers, the propellant charge molded bodies forming this layer are each designed in such a way that they are adapted in their entirety to the shape of the propellant chamber. In other words, the propellant charge molded bodies of a layer are overall shaped to fit the shape of the propellant chamber. This contributes to a comparatively high charge density in the propellant chamber and thus to a high charge mass. Adapted to the shape of the propellant chamber means that the propellant charge molded bodies forming a layer are (shape-)adapted as a whole to the shape or contour of the propellant chamber or the shell in such a way that as few empty spaces as possible remain in the spaces between the propellant charge molded bodies and/or between the propellant charge molded bodies and the shell (adapted to the contour of the shell). In particular in combination with an increased number of holes (at least 70 holes per propellant charge molded body), the shape adjustment can increase the charge mass having the same propellant chamber volume by approx. 20 to 40 percent of the charge density of approx. 1 g/cm.³ currently achieved with 19-hole molded bodies.

[0021] Conveniently, each propellant charge molded body can have a cross section (base area) and can extend with this cross section along a molded body axis. This facilitates a structurally simple and robust design. The propellant charge molded body may be a cylindrical body. The molded body axis can be oriented orthogonally to the base surface. The holes can be arranged parallel, orthogonally or obliquely (skewed) to the molded body axis.

[0022] The cross section of the propellant charge molded body can be circular, partially circular (e.g., circular sector or circular segment) or polygonal (n-polygonal), wherein the polygonal design can have a regular or irregular geometry. For an n-polygonal design, $3 \leq n \leq 8$ or $3 \leq n \leq 6$ is possible. In the case of an n-polygonal design, a body side of the molded body extending parallel to the molded body axis (body side on the lateral surface) can be convex or concave or shaped convexly or concavely.

[0023] Specifically, the propellant charge molded body may have a hexagonal cross section and at least 60 holes, e.g., 61 holes.

[0024] The propellant charge molded bodies can be adapted to the shape or geometry of the propellant chamber by means of bevels, recesses and/or slots.

[0025] Further, the propellant charge molded bodies can have an identical or substantially identical cross section in shape and/or size.

[0026] For example, the propellant charge molded bodies having a circular cylindrical propellant chamber or circular cylindrical shell can have a cross section in the form of a circular sector (“pie slice”). One or more layers of the propellant charge can each be formed from at least two such propellant charge molded bodies.

[0027] Advantageously, however, the propellant charge molded bodies can have cross sections that differ in shape and/or size. This can be used to increase the charge density in the propellant chamber if necessary. In addition, the assembly of the propellant charge molded bodies can be adapted to other geometric elements in the propellant chamber, e.g., to projectile components (bullet, e.g., penetrator, and/or sabot) that protrude into the propellant chamber. Specifically, the propellant charge may contain different propellant charge molded bodies, each having a cross section that differs in shape and/or size, for example two, three, four or more cross sections that differ in shape and/or size.

[0028] The holes can conveniently be designed as blind holes and/or through holes. In other words, one portion of the holes can be designed as blind holes and another portion as through holes. Alternatively, all holes can be designed as blind holes or through holes. Holes can be oriented parallel to the molded body axis, orthogonally to the molded body axis or obliquely to the molded body axis. The holes can be oriented parallel to one another (perforation). Through holes can be produced comparatively easily in the propellant charge molded body. The increase in surface area (progressive combustion) can be further increased by means of blind holes.

[0029] The predominant portion of the holes, in particular at least 60, 70 or 80 percent of the holes, on the at least one propellant charge molded body can be arranged with the same hole spacing from one another (holes arranged equidistant from one another), wherein the remaining holes are arranged with a different hole spacing from one another, in particular with less distance between the holes. If deviations from the same (equidistant) spacing are allowed for the remaining holes, the holes on the propellant charge molded body can be arranged much more variably, which promotes a high degree of variability in the shape of the propellant charge molded body. In addition, the number of holes can be increased while maintaining the same shape and/or size of a propellant charge molded body. A non-optimal (non-equidistant) choice of spacing between the holes only insignificantly affects the overall increase in surface area during combustion.

[0030] Advantageously, the propellant charge molded bodies can be (surface-) treated in such a way that the holes are partially closed by a substance (e.g., a solid) which reduces, preferably compensates, the temperature dependence of the combustion rate of the propellant charge molded body. The propellant charge molded bodies treated in this way burn almost independently of their initial temperature. Closing the holes with the substance can counteract the increase in the combustion rate of the propellant powder formulation with the temperature, so that the temperature dependence of the gas pressure development is partially compensated. This behavior allows a nearly constant output power across the entire temperature range. Due to the lower gas pressure level at high temperatures—and the limitation of the weapon stress associated with the gas pressure level (upper gas pressure limit)—the necessary performance can also be achieved at low temperatures.

[0031] Exemplary, one of the propellant charge molded bodies can have a diameter of 50 mm (millimeters) and a height of 35 mm. This propellant charge molded body can, for example, have 470 holes and a hole spacing (so-called “web”) of, for example, 2 millimeters (partial deviations from the ideal hole spacing are possible, as explained above). The holes can have a diameter of 0.2 mm to 0.5 mm.

[0032] The shell delimiting the propellant chamber may have a hollow cylindrical shape. The shell may optionally be closed at its axial ends (closed hollow cylinder).

[0033] The shell which outwardly delimits the propellant chamber may be made of paper (paper shell), preferably of such paper which also burns when the propellant charge powder burns, in particular without leaving any residue. The shell which outwardly delimits the propellant chamber can alternatively be made, for example, from a mixture of wood fibers, nitrocellulose, stabilizers (for the chemical stability of the nitrocellulose) and a composite material (varnish or adhesive), so that the shell is sufficiently stable for the application and also burns, in particular without leaving any residue.

[0034] Alternatively, the shell which outwardly delimits the propellant chamber can be made of a textile fabric (fabric shell). The fabric can preferably be designed in such a way that it also burns when the propellant powder burns, in particular without leaving any residue. The fabric cover can be designed as a bag, for example.

[0035] The propellant charge may also be formed of a plurality of bags.

[0036] The shell that outwardly delimits the propellant chamber can be made of metal (metal shell). Such a shell forms a solid and stable boundary of the propellant chamber. This shell can be designed, for example, as a propellant charge shell or as a cartridge shell.

[0037] The aforementioned object is also achieved by a propellant charge assembly having features of the independent claim. With regard to the advantages that can be achieved with the propellant charge assembly, reference is made to the statements relating to the propellant charge in this respect.

[0038] The propellant charge assembly comprises one or more propellant charges having one or more of the aspects described above.

[0039] For example, the propellant charge assembly may comprise one or more separately encased propellant charges, depending on the desired power to drive the projectile. The individual propellant charges can be enclosed in bags and/or positioned relative to one another by appropriate holders, in particular those made of combustible material. This allows for targeted loading of the propellant chamber of a barrel weapon, for example as in the case of a howitzer.

[0040] The aspects mentioned above in connection with the propellant charge can serve for developing the propellant charge assembly.

[0041] The object mentioned at the outset is also achieved by a piece of ammunition having the features of the independent claim. With regard to the advantages that can be achieved thereby, reference is made to the statements made in this respect about the propellant charge.

[0042] The ammunition may be designed as cartridge ammunition within the scope of a possible configuration. Alternatively, the ammunition may also be constructed in two parts, with a projectile part possibly containing propellant powder and a pure propellant powder part.

[0043] The piece of ammunition comprises a projectile and a propellant charge having at least one of the aspects described above. The shell, which outwardly delimits the propellant chamber, can in this case be designed as a cartridge shell, for example as a cartridge shell made of metal. In this way, particularly powerful cartridge ammunition can be provided.

[0044] Advantageously, the projectile may comprise a projectile body, in particular a penetrator. Due to the high gas mass, the projectile body can reach a comparatively high muzzle velocity ($v_{sub.0}$) and achieve high penetration performance at the target.

[0045] Preferably, the projectile (in addition to the projectile body) may comprise a sabot for the projectile body. The sabot is a guidance device that guides the sub-caliber projectile body in the (larger diameter or “full caliber”) gun barrel and serves to seal the gun barrel against propellant gases. The seal allows the gas pressure generated when the propellant charge burns to be transferred to the projectile body. After leaving the muzzle of the gun barrel, the sabot falls off the projectile body.

[0046] Due to the more progressive combustion of the propellant charge molded bodies proposed here, the projectiles are subjected to less stress by reducing the maximum gas pressure. This allows less solid and lighter materials to be used for propellant bases, e.g., plastics or pressed structural materials for the sabot.

[0047] When designed as cartridge ammunition, the ammunition may be constructed in such a way that the cartridge shell has a shell base on or above which (toward the propellant chamber of the propellant charge) a propellant charge igniter is arranged. The propellant igniter can be a flame, an electrical igniter or a pyrotechnic composition. The propellant charge (located in the propellant chamber) is adjacent to the propellant charge igniter.

[0048] The object mentioned at the outset is also achieved by a set formed of or consisting of a barrel weapon, a projectile and a propellant charge to drive the projectile. With regard to the advantages that can be achieved thereby, reference is made to the statements made in this respect about the propellant charge.

[0049] The barrel weapon has a gun barrel and a propellant chamber adjacent to the gun barrel. The propellant chamber is located in particular “upstream” of the gun barrel, i.e., at the end of the gun barrel facing away from the muzzle. The projectile may be inserted into the gun barrel and “applied” thereto, for example in a howitzer. The propellant chamber of the barrel weapon serves to receive the propellant charge.

[0050] The propellant chamber has a propellant chamber wall that outwardly limits the propellant chamber. At least two, or more than two, propellant charge molded bodies are arranged in the propellant chamber, wherein the propellant charge molded bodies each have a plurality of holes extending next to one another. The holes may preferably extend parallel to one another. At least one of the propellant charge molded bodies, preferably two, or more than two, propellant charge molded bodies, each have at least 70 holes extending next to one another.

[0051] Instead of the propellant charge shell which outwardly delimits the propellant chamber, the propellant chamber into which the propellant charge molded bodies are or were introduced is delimited by the propellant chamber wall. In other words, the shell of the propellant charge, which outwardly delimits the propellant chamber, can be omitted.

[0052] The propellant charge molded bodies may be used not only in cartridge ammunition, but also without a shell or in bulk or in one or more bags.

[0053] Advantageously, the propellant charge molded bodies may be arranged or layered on top of one another in the propellant chamber of the barrel weapon in one, two or more layers, wherein at least two propellant charge molded bodies each form a layer, wherein in at least one of the layers, preferably two, a plurality of or all of the layers, the propellant charge molded bodies forming this layer are each designed in such a way that they are adapted in their entirety to the shape of the propellant chamber of the barrel weapon.

[0054] The measures described in connection with the propellant charge may be used to further develop the propellant charge molded body of the set.

[0055] Further scope of applicability of the present invention will become apparent from the detailed description given hereinafter. However, it should be understood that the detailed description and specific examples, while indicating preferred embodiments of the invention, are given by way of illustration only, since various changes, combinations, and modifications within the spirit and scope of the invention will become apparent to those skilled in the art from this detailed description.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0056] The present invention will become more fully understood from the detailed description given hereinbelow and the accompanying drawings which are given by way of illustration only, and thus, are not limitive of the present invention, and wherein:

[0057] FIG. 1 shows a longitudinal section through an example of a cartridge ammunition;

[0058] FIG. 2 shows a cross section through an example of a cartridge ammunition;

[0059] FIG. 3 shows an example of the cartridge ammunition from FIG. 2 in a cross section;

[0060] FIG. 4 shows an example of the cartridge ammunition from FIG. 2 in a cross section;

[0061] FIG. 5 shows a cross section through an example of a cartridge ammunition;

[0062] FIG. 6 shows a cross section through an example of a cartridge ammunition;

[0063] FIG. 7 shows a cross section through an example of a cartridge ammunition;

[0064] FIG. 8 shows a cuboid-shaped propellant charge molded body in a perspective view and a sectional view;

[0065] FIG. 9 shows an example of the cuboid-shaped propellant charge molded body from FIG. 8 with a convex-molded body side in a perspective view and a sectional view;

[0066] FIG. 10 shows a longitudinal section through a cuboid-shaped propellant charge molded body with through holes; and

[0067] FIG. 11 shows a longitudinal section through a propellant charge molded body with triangular cross section and through holes oriented transversely to the molded body axis.

DETAILED DESCRIPTION

[0068] FIG. 1 shows a schematic longitudinal section of a cartridge ammunition which is designated as a whole by the reference sign **300**.

[0069] The cartridge ammunition **300** has a projectile **302** with a projectile body **304** and a sabot **306** for the projectile body **304**. In the example, the projectile body **304** is designed as a penetrator having a main body **308** and a tail unit **310** attached to the penetrator rear. The cartridge ammunition **300** also includes a propellant charge **100** for driving the projectile **302** through a gun barrel of a barrel weapon.

[0070] The propellant charge **100** has a shell **102** which outwardly delimits a propellant chamber **103**. In the example, the shell **102** is designed as a solid shell **104** (propellant charge or cartridge shell **104**), preferably made of combustible shell material. A sleeve base **106** (shown only schematically here) is attached to the sleeve **104**, which closes the sleeve **104** at the rear. A propellant charge igniter may be mounted on or above the shell base **106**.

[0071] A plurality of propellant charge molded bodies **110** is arranged in the propellant chamber **103**, wherein the propellant charge molded bodies **110** each have a plurality of holes **112** extending next to one another (holes are only indicated for reasons of clarity). The holes **112** may preferably extend parallel to one another. In the example, at least half of the propellant charge molded bodies **110** each have at least 70 holes **112** extending side by side. The propellant charge molded bodies in question can have 100 or more holes **112**, as explained above. The holes **112** can be designed as blind holes or as through holes.

[0072] Each propellant charge molded body **110** has a cross section (base area) and extends with this cross section along a molded body axis **114** (shown on one of the propellant charge molded bodies **110**). The cross section of a propellant charge molded body **110** may be circular, partially circular or polygonal, as explained above. The propellant charge molded bodies **110** may have a different or identical cross section. Irrespective of this, the holes **112** in the example are oriented parallel to the molded body axis **114**.

[0073] The propellant charge molded bodies **110** can each have the same height or are designed to be the same height. In the example, the propellant charge molded bodies **110** have different heights (different axial structures) to illustrate different designs. Thus, the propellant charge molded bodies **110** arranged on the right in FIG. 1 are approximately twice as high as the propellant charge molded bodies **110** arranged on the left in FIG. 1.

[0074] The propellant charge molded bodies **110** are arranged in the propellant chamber **103** in a plurality of layers S, wherein at least two propellant charge molded bodies **110** each form a layer S (the layers S' and S'' are marked in FIG. 1). In these layers S', S'' (marked in FIG. 1), the this layer S', S'' are each formed in such a way that they are adapted in their entirety to the shape of the propellant chamber **103**. In other words, the propellant charge molded bodies **110**, each forming a layer S', S', are adapted as best as possible to the shape of the propellant chamber **103**.

[0075] Optionally, on the propellant charge molded bodies **110**, which have at least 70 or at least 100 holes **112**, the majority of the holes **112**, for example more than 50 percent of the holes **112**, can be arranged with the same hole spacing from one another (holes equidistant), wherein the remaining holes **112** can be arranged with a smaller hole spacing from one another, for example, as explained above.

[0076] Irrespective of this, the propellant charge molded bodies can be surface treated, as explained above.

[0077] FIG. 2 shows a cross section through an example of a cartridge ammunition **300**, wherein, for the sake of illustration only, one propellant charge molded body **110** is shown in the propellant chamber **103**.

[0078] The propellant charge molded body **110** can be designed as described above and have at least 70 or at least 100 holes **112** (holes only partially indicated in FIG. 2). The propellant charge molded body **110** has a cross section in the form of a circular sector ("pie slice") having a flattened tip in the example, wherein the propellant charge molded body **110** with this cross section extends

along the molded body axis **114** (molded body axis **114** protrudes orthogonally from the plane of the drawing in FIG. 2).

[0079] In the example, the propellant charge molded body **110** rests as closely as possible with its radially outer surface portion **120** against the sleeve **104** and with its radially inner surface portion **122** against the main body **308**. As a result, the propellant chamber **103** between the main body **308** and the sleeve **104** is largely filled (proportionately based on the circular sector or the relevant propellant charge molded body **110**).

[0080] In the example, six identically formed propellant charge molded bodies **110** form a layer of the propellant charge **100** (indicated by dashed lines in FIG. 2), wherein the propellant charge molded bodies **110** are formed with the same thickness along the molded body axis **114**. The propellant charge molded bodies **119**, which in the example form a layer S, are designed in such a way that they are adapted as a whole to the shape of the propellant chamber **103** (as few empty spaces as possible between the molded bodies **110** and/or the shell **104**).

[0081] In the example, a plurality of layers S of propellant charge molded bodies **112** can be provided, in particular more than two layers.

[0082] FIG. 3 shows a possible design that largely corresponds to the cartridge ammunition **300** from FIG. 2. To avoid repetition, reference is therefore made to the statements above regarding FIG. 2.

[0083] By contrast, the cartridge ammunition according to FIG. 3 has a plurality of fins **312** projecting radially outwardly from the main body **308**, which fins form, for example, a tail unit **310** of the penetrator. The two propellant charge molded bodies **110** arranged in the propellant chamber **103** for illustration purposes each have a cross section in the form of a circular sector with a flattened tip in the example, wherein these propellant charge molded bodies **110** with this cross section each extend along their molded body axis **114**.

[0084] The propellant charge molded bodies **110** are dimensioned such that they each fit into the space between two adjacent fins **312**. An empty space (slot) remaining between two adjacent propellant charge molded bodies **110** can optionally be filled, for example, with propellant charge powder having a correspondingly fine grain size (rod or bulk powder). The propellant charge molded bodies **110** are identical in the example.

[0085] FIG. 4 shows another possible design, which largely corresponds to the cartridge ammunition **300** from FIG. 2. To avoid repetition, reference is therefore made to the statements above regarding FIG. 2.

[0086] The propellant charge molded bodies **110** arranged between the main body **308** and the sleeve **104** also have overall a cross section in the form of a circular sector ("pie slice") with a flattened tip in the example. However, these propellant charge molded bodies **110** are in the present case formed in a plurality of parts and each have a radially inner molded body part **110'** and a radially outer molded body part **110''**. The radially inner molded body part **110'** and the radially outer molded body part **110''** adjoin one another at a parting plane T, wherein the parting plane T in the example is oriented orthogonally to the radial direction (other, e.g., oblique, orientation of the parting plane T is also possible).

[0087] The radially outer molded body part **110''** has a cross section in the form of a circular sector with, in the example, a further flattened tip (parting plane T). In other words, the radially outer molded body part **110''** has the cross section of a circular segment. The radially inner molded body part **110'** has a cross section in the form of an isosceles trapezoid. The radially inner molded body part **110'** and the radially outer molded body part **110''** each extend with their cross section along their molded body axis **114**. Two different propellant charge molded bodies or propellant charge molded body parts are arranged in the propellant chamber **103** by the two molded body parts **110'**, **110''**. Empty spaces in the propellant chamber **103**, for example between two molded body parts **110'**, **110''** and/or the sleeve **104**, can optionally be filled, for example, with propellant powder having a correspondingly fine grain size (loose or rod powder).

[0088] FIG. 5 shows a cross section through an example of a cartridge ammunition in which different propellant charge molded bodies are arranged in the propellant chamber **103**.

[0089] First, a multi-part propellant charge molded body **110** with a radially inner molded body part **110'** and a radially outer molded body part **110''** is arranged in the propellant chamber **103**. To avoid repetition, reference is therefore made to the statements above regarding FIG. 4.

[0090] In addition, two propellant charge molded bodies **124** are arranged in the propellant chamber **103**, each of which has a circular cross section and extends with this cross section along its molded body axis **114**. In addition, in the example a further propellant charge molded body **124'** with a circular cross section is arranged in the propellant chamber **103**, which has a significantly smaller diameter than the two propellant charge molded bodies **124**. The propellant charge molded body **124'** is arranged between the two propellant charge molded bodies **124** and the sleeve **104**. The propellant charge molded body **124'** thus serves to increase the charge density (filling gaps).

[0091] Furthermore, a plurality of, in this case only eight, propellant charge molded bodies **124''** having a circular cross section are arranged in the propellant chamber **103**, which in this example have an even smaller diameter than the propellant charge molded body **124'**.

[0092] At least the propellant charge molded bodies **110** and **124**, preferably all propellant charge molded bodies **110**, **124**, **124'**, **124''**, each have at least 70 or at least 100 holes **112** extending next to one another.

[0093] In the present case, five different propellant charge molded bodies **110**, **124** are used to fill the propellant chamber **103** or to fill intermediate spaces, so that the highest possible charge densities can be achieved. Remaining empty spaces in the propellant chamber **103**, e.g., empty spaces between the different propellant charge molded bodies **110**, **124** and/or the shell **104**, can optionally be filled, e.g., with propellant charge powder having a correspondingly fine grain size (loose or stick powder).

[0094] FIG. 6 shows a cross section through an example of a cartridge ammunition in which a plurality of identical propellant charge molded bodies **126** are arranged in the propellant chamber **103**.

[0095] A plurality of propellant charge molded bodies **126** are arranged in the propellant chamber **103**, each of which has a preferably regular six-sided or hexagonal cross section and which extends with this cross section along its molded body axis **114**. The propellant charge molded bodies **126** each have at least **70** or at least **100** holes **112** extending next to one another.

[0096] In the example here, a plurality of, in this case **18**, propellant charge molded bodies **126** with hexagonal cross section are arranged directly next to one another and form a layer **S** of the propellant charge **100**.

[0097] The hexagonal propellant charge molded bodies **126** are adapted to the geometry of the shell **104** or the propellant chamber **103** in order to achieve the highest possible charge density. The holes **112** are made in the propellant charge molded bodies **126** with corresponding spacing, depending on the maximum gas pressure, projectile mass, etc. Remaining empty spaces in the propellant chamber **103**, e.g., empty spaces between the propellant charge molded bodies **126** and/or the shell **104**, can optionally be filled, e.g., with propellant charge powder having a correspondingly fine grain size (loose or rod powder). The increase in charge mass can be converted into an increase in the speed of the projectile.

[0098] FIG. 7 shows a cross section through an example of a cartridge ammunition **300**, in which various propellant charge molded bodies are arranged in the propellant chamber **103**.

[0099] First, a multi-part propellant charge molded body **110** with a radially inner molded body part **110'** and a radially outer molded body part **110''** is arranged in the propellant chamber **103**. To avoid repetition, reference is therefore made to the statements above regarding FIG. 4.

[0100] In addition, two propellant charge molded bodies **124** are arranged in the propellant chamber **103**, each of which has a circular cross section and extends with this cross section along its molded body axis **114**. To avoid repetition, reference is therefore made to the statements above

regarding FIG. 5.

[0101] Furthermore, arranged in the propellant chamber **103** is a propellant charge molded body **126**, which has a six-sided or hexagonal cross section and extends with this cross section along its molded body axis **114**. To avoid repetition, reference is therefore made to the statements made above in this regard in relation to FIG. 6.

[0102] Furthermore, a propellant charge molded body **128** is arranged in the propellant chamber **103**, which has a triangular cross section and extends with this cross section along its molded body axis **114**. In the example, the propellant charge molded body **128** is arranged between the two propellant charge molded bodies **124** and the sleeve **104**.

[0103] The propellant charge molded bodies **110**, **124**, **126**, **128** can each have at least 70 or at least 100 holes **112**, as explained above, wherein the holes of these propellant charge molded bodies are each oriented parallel to the relevant molded body axis **114**. The holes **112** can be designed as blind holes or as through holes.

[0104] The majority of the holes **112**, for example more than 50 percent of the holes **112**, of the propellant charge molded bodies **110**, **124**, **126**, **128** can be arranged with the same hole spacing from one another (holes equidistant), wherein the remaining holes **112** on the relevant propellant charge molded body **110**, **124**, **126**, **128** can be arranged with a smaller hole spacing from one another, for example, as explained above.

[0105] In the propellant chamber **103**, there are also arranged a plurality of, in this case by way of example four, propellant charge molded bodies **130** having a cuboid shape, each of which extends along its molded body axis **114**. In addition, a plurality of, in this case by way of example three, propellant charge molded bodies **132** having a cuboid shape with a convexly shaped body side are arranged in the propellant chamber **103**, each of which extends along its molded body axis **114**. The propellant charge molded bodies **130**, **132** are described below.

[0106] FIG. 8 shows the propellant charge molded body **130** in a perspective view and in a sectional view. The propellant charge molded body **130** can have at least 70 or 100 holes **112**, as explained above, wherein the holes **112** of the propellant charge molded body **130** are each oriented transversely or orthogonally to the molded body axis **114**.

[0107] The holes **112** can be designed as blind holes or as through holes. The holes **112**, which are designed as blind holes, can be formed from two opposite flat sides of the propellant charge molded body **130**. Thus, a portion of the blind holes may be formed from a first flat side **130'** and another portion of the blind holes may be formed from a second flat side **130''**. Alternatively, the blind holes may all be formed starting from only one of the flat sides **130'**, **130''**.

[0108] Further, the holes **112** can alternatively be oriented parallel or obliquely to the molded body axis **114**.

[0109] FIG. 9 shows the propellant charge molded body **132** in a perspective view and in a sectional view.

[0110] The propellant charge molded body **132** corresponds in its design to the propellant charge molded body **130**, so that, to avoid repetition, reference is made to the explanations there.

[0111] Deviating from this, the propellant charge molded body **132** is convex on a body side **132''** extending parallel to the molded body axis **114**. The opposite body side **132'** is flat in the example. This can also be used to influence the charge density of the propellant chamber **103**.

[0112] FIG. 10 shows a schematic longitudinal section of another propellant charge molded body **130** which is plane parallel (first flat side **130'** and second flat side **130''** are oriented parallel to one another). The present propellant charge molded body may have a rectangular cross section (shown here) or alternatively a circular cross section. In the present propellant charge molded body **130**, all holes **112** are designed as through holes. The propellant charge molded body **130** can, for example, contribute to a high charge density in the propellant chamber due to the comparatively low height (extension along the molded body axis **114**).

[0113] FIG. 11 shows a schematic longitudinal section of a further propellant charge molded body

134 which extends, at least part of the way, along the molded body axis **114**. The first body side **134'** is oriented orthogonally to the molded body axis **114**, and the second body side **134''** is angled relative to the first body side **134'**. In other words, the two body sides **134'**, **134''** form an angle α with one another, wherein in the example α is $<45^\circ$. In the example, the propellant charge molded body **134** has the cross section of a circular sector (circular arc in FIG. **11** left). In the present case, all holes are designed as through holes **112**, which extend from the first body side **134'** to the second body side **134''**.

[0114] By means of the propellant charge molded body **134**, a conically tapering portion of the propellant chamber (e.g., a “dome”) can be filled with a high charge density. Figuratively speaking, the propellant charge molded body **134** resembles a pie slice which is very thin at the edge (in FIG. **11** left) and very high in the middle (in FIG. **11** right).

[0115] The invention being thus described, it will be obvious that the same may be varied in many ways. Such variations are not to be regarded as a departure from the spirit and scope of the invention, and all such modifications as would be obvious to one skilled in the art are to be included within the scope of the following claims.

Claims

1. A propellant charge for driving a projectile, the propellant charge comprising: a shell that outwardly delimits a propellant chamber; and at least two propellant charge molded bodies, which are arranged in the propellant chamber, each of the at least two propellant charge molded bodies having a plurality of holes extending next to one another, wherein at least one of the propellant charge molded bodies has at least 60 respective holes extending next to one another.
2. The propellant charge according to claim 1, wherein the at least one propellant charge molded body has at least 100, at least 200, at least 300, or at least 400 holes extending next to one another.
3. The propellant charge according to claim 1, wherein the propellant charge molded bodies are arranged on top of one another in the propellant chamber to form one, two, or more layers, wherein at least two propellant charge molded bodies each form a layer, wherein at least in one layer, in two layers, multiple layers, or all layers, the propellant charge molded bodies forming this layer are designed such that they are adapted in their entirety to the shape of the propellant chamber.
4. The propellant charge according to claim 1, wherein each propellant charge molded body has a cross section and extends with this cross section along a molded body axis.
5. The propellant charge according to claim 4, wherein the propellant charge molded bodies have cross sections that differ in shape and/or size.
6. The propellant charge according to claim 1, wherein the holes are designed as blind holes and/or through holes.
7. The propellant charge according to claim 1, wherein on the at least one propellant charge molded body, the holes are predominantly arranged with the same hole spacing from one another, and wherein the remaining holes are arranged with a different hole spacing from one another or with a smaller hole spacing.
8. The propellant charge according to claim 1, wherein the propellant charge molded bodies are treated such that the holes are partially closed by a substance which reduces the temperature dependence of the combustion rate of the propellant charge molded body.
9. A propellant charge assembly comprising one or more propellant charges according to claim 1.
10. An ammunition comprising a projectile and the propellant charge according to claim 1.
11. The ammunition according to claim 10, wherein the projectile comprises a projectile body or a penetrator.
12. The ammunition according to claim 11, wherein the projectile has a sabot for the projectile body.
13. A set comprising: a barrel weapon; a projectile; and a propellant charge for driving the

projectile, wherein the barrel weapon has a gun barrel and a propellant chamber adjacent to the weapon barrel, which is outwardly delimited by a chamber wall, wherein at least two propellant charge molded bodies are arranged in the chamber, each of the at least two propellant charge molded bodies having a plurality of holes extending next to one another, and wherein at least one of the propellant charge molded bodies has at least 70 holes extending next to one another.
